

A Parameter Refinement of the Seven-Eighths Zero-Free Half-Plane

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Abstract

We refine the parameter choices in OpenAI’s claimed zero-free half-plane $\operatorname{Re} s > 7/8$ for finite-order Hecke L -functions over $\mathbb{Q}(\sqrt{-3})$. Here s is a complex number and $\operatorname{Re} s$ is its real part, so the claimed region consists of those s whose real part is greater than $7/8$. Taking the source’s arithmetic and analytic estimates as inputs, we obtain the sharper boundary

$$\sigma_* = 0.874957019420098946\dots, \quad 7884\sigma_*^3 - 18819\sigma_*^2 + 14643\sigma_* - 3686 = 0.$$

Since $\sigma_* < 162861/186136 = 7/8 - 1/23267$, the half-plane $\operatorname{Re} s > 162861/186136$ is zero-free as well. The argument retains the exact zero-free-parameter dependence in a selected-prime fourth-moment bound, extends that bound to the required parameter range, and optimizes the geometry of the existing compensated probe. We verify the resulting endpoint inequality and the affected Euler-product and contour estimates, obtaining a contracting bootstrap map with fixed point σ_* . The conclusion transfers, as in the source, to all Dirichlet L -functions. This is a *source-relative* refinement: it neither reproves nor independently validates the deep estimates in the cited manuscript.

Keywords. Zero-free regions; Dirichlet L -functions; Hecke L -functions; compensated probes; moment estimates.

1 Introduction and main result

The compensated-probe method of [1] is reported to give the uniform zero-free region $\operatorname{Re} s > 7/8$ for finite-order Hecke L -functions over $\mathbb{Q}(\sqrt{-3})$, with a transfer to Dirichlet L -functions. Its main input is a pair of estimates for a finite arithmetic probe: a direct bound and a Mellin representation whose principal signal contains a reciprocal L -function. A zero to the right of the permitted boundary would contradict the continuation obtained from these estimates.

The purpose of this note is narrower than establishing those inputs. We preserve the source’s finite probe and its deep reflection, detector, and moment identities, but sharpen the treatment of the selected-prime capacity and optimize the allowed asymmetric scales. A parameter-dependent fourth-moment estimate, an exact endpoint inequality, and an audit of the affected Euler and contour regions make the resulting improvement iterable. The gain in the boundary is small, but strict.

Let $F = \mathbb{Q}(\sqrt{-3})$. Write β_* for the supremum of $1/2$ and of the real parts of zeros in $1/2 \leq \operatorname{Re} s \leq 1$ of all primitive finite-order Hecke L -functions over F . Poles are excluded. The source [1] proves $\beta_* \leq 7/8$. Its transfer proposition passes a bound on this family to all finite-order Hecke functions and all Dirichlet L -functions, including ζ .

Theorem 1.1 (Source-relative refinement). *Assume the results of [1] listed below, with their stated uniformity and coefficient conventions. Let σ_* be the unique root in $(17499/20000, 7/8)$ of*

$$7884s^3 - 18819s^2 + 14643s - 3686 = 0. \quad (1.1)$$

Then $\beta_ \leq \sigma_*$. In particular every finite-order Hecke L -function over F , and every Dirichlet L -function, is zero-free in the strict half-plane $\operatorname{Re} s > \sigma_*$. A principal pole at one is allowed. No assertion of nonvanishing on $\operatorname{Re} s = \sigma_*$ is made. Since*

$$\sigma_* < \frac{162861}{186136} = \frac{7}{8} - \frac{1}{23267}, \quad (1.2)$$

these functions are, in particular, zero-free in $\operatorname{Re} s > 162861/186136$.

Dependence on the source. We use the 7/8 theorem and the exact compensated probe and Poisson identity from [1]; its cubic-theta reflection, finite-correlation and additive Gram estimates; its buffered zero detector, strict marked inverse moment and sixth-power amplification; and the parameter-independent parts of its plain fourth-moment argument. We also invoke the source's logarithmic control, fixed-ray prime normalization, contour accounting, principal-residue, outer-tail, late-height and transfer results. The changes to their relevant parameter ranges are justified in the appendices. In particular, no uniform theorem for arbitrary moving coefficients or punctures is taken as an additional input.

Source equation and lemma labels printed in typewriter font refer to the frozen version of [1] identified in the bibliography.

2 The improvement map and compensated probe

2.1 Parameters and bootstrap

For an already known upper bound

$$\beta_* \leq \sigma, \quad \frac{17499}{20000} \leq \sigma \leq \frac{7}{8}, \quad \kappa = 2\sigma - 1,$$

define

$$d_\kappa = \frac{156\kappa - 19 - \sqrt{9936\kappa^2 - 2808\kappa + 201}}{12(12\kappa - 1)}, \quad (2.1)$$

$$\ell = \frac{5 - 6d_\kappa}{9 + 18d_\kappa}, \quad (2.2)$$

$$b = (1 + 3\ell) \frac{36\kappa d_\kappa - 6\kappa - 1}{(156\kappa - 19) - 12(12\kappa - 1)d_\kappa}, \quad (2.3)$$

$$l_x = \frac{1 - \ell - b}{2}, \quad l_y = \frac{1 - \ell + b}{2}, \quad h = \frac{1 + 3\ell + b}{2}. \quad (2.4)$$

Thus $l_x + l_y + \ell = 1$ and $h = 1 - l_x + \ell$. The symbol d_κ is one critical zero-bin parameter; it is not the row-scale exponent d used in the estimates below. The improvement map is

$$f(\sigma) = \frac{11}{12} - \frac{\ell}{4} = 1 - \frac{2}{9 + 18d_{2\sigma-1}}. \quad (2.5)$$

Proposition 2.1 (Bootstrap step). *Under the source hypotheses of Theorem 1.1, for every $\sigma \in [17499/20000, 7/8]$,*

$$\beta_* \leq \sigma \implies \beta_* \leq f(\sigma). \quad (2.6)$$

The fixed-point calculation in Appendix B shows that f has a unique fixed point σ_* in this interval and satisfies $f(\sigma) < \sigma$ above that point. Thus Proposition 2.1, iterated from 7/8, implies Theorem 1.1.

2.2 The unchanged arithmetic probe

For clarity, we retain the source's finite arithmetic observable rather than defining a new probe by its desired Mellin representation. Put $X = Z^{l_x}$, $Y = Z^{l_y}$, and choose disjoint fixed-ray prime windows of lengths $P_i = Z^{\ell_i}$, where $\sum_i \ell_i = \ell$. Write $I_{\eta;D}$ for the source's completed cubic-theta probe with $1_{D|cn^3}$ inserted. For $p_J = \prod_{i \in J} p_i$, the physical probe is exactly

$$I_{\eta, \text{modified}}(Z) = \sum_{(p_i)} \prod_i W_i(q_{p_i}/P_i) \sum_{J \subseteq \{1, \dots, K\}} (-1)^{|J|} q_{p_J}^{-3/2} \overline{\eta(p_{J^c})} \cdot I_{\eta; p_{J^c}}(X/q_{p_J}, Y/q_{p_J}, Zq_{p_{J^c}}). \quad (2.7)$$

Here $q_a = Na$, every slot retains its original window under all rescalings, and all source zero extensions and fixed-ray phases remain. Only its real exponents and the number of fixed slots are changed. This finite expression is the source's **eq:compensated-probe-definition**.

At a selected prime its two terms cancel the scalar local contribution, leaving the character prime factor to which the inverse and plain moments apply. The gain proved here comes from keeping the exact prime capacity and changing the geometry around that cancellation. No new off-diagonal cancellation estimate is assumed.

2.3 Proof structure

The refinement has three inputs beyond the assertions imported from [1]. Appendix A extends the plain fourth moment and derives a selected-row count without the source's fixed-capacity comparison loss. Appendix B optimizes the geometry and proves the global endpoint inequality by rational coefficient bounds; its fixed-point calculation identifies σ_* . Appendix C verifies the perturbed low estimate, Euler-product region, contour inequalities, and Mellin continuation with the needed quantifier order. The bootstrap is assembled in Section 3.

3 Proof of the refinement

Proof of Proposition 2.1 and Theorem 1.1. Fix an input $\sigma \in [17499/20000, 7/8]$ with $\beta_* \leq \sigma$, and take (2.1)–(2.4). The estimates in Appendices B and C establish the needed geometric and Euler inequalities. Suppose, for a contradiction, that

$$\Delta = \beta_* - f(\sigma) > 0.$$

This is a fixed positive number for the entire Hecke family. It need not be effective or known numerically, and it does not depend on the target character.

The extended fourth moment from Appendix A applies because $\beta_* \leq (1 + \kappa)/2$. The buffered dynamic bins satisfy $\delta \leq 2\beta_* - 1 \leq \kappa$. The exact selected-row bound therefore applies with the same κ and gives the envelope (A.18); it is not permissible to insert the desired new bound in place of this already available input. The global algebraic inequality gives $E(h) \leq 0$. The floor, outer, and principal contributions have the explicit strict margins checked above.

Choose the positive moment losses, bin and amplitude widths, capacity decrements, and row extension so that their combined real cost is smaller than a fixed fraction of Δ and of the other strict margins. Choose a sufficiently fine fixed prime mesh and an even number of disjoint slot windows. Their existence uses the positive supply margin, and their normalizer is nonzero for all sufficiently large Z . Every one of these real choices precedes the target character. The source's coefficient classes remain intact because each selected prime coefficient is a fixed finite sum of fixed-ray characters, independent of the varying row.

For the chosen target η , fix its arithmetic data and excluded set. Fix the finite internal seminorm and height orders first, then choose the height exponent small enough to absorb

them, and only afterwards choose the external tail order. The late-height argument in the analytic audit supplies a positive saving independent of η . With the same scalar normalizer on both sides, it follows that

$$|J_\eta(Z)| \ll_\eta Z^{C(f(\sigma))+\Delta/2}, \quad |J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*)-m}$$

for one $m > 0$ independent of η . The notation f_η denotes the Mellin signal, whereas $f(\sigma)$ denotes the numerical improvement map. These are different objects.

The Euler multiplier of the signal is holomorphic and bounded away from zero on $\operatorname{Re} s > f(\sigma)$. Its Mellin transform consequently continues every reciprocal $1/L_F^S(s, \eta)$ into a common half-plane strictly to the left of β_* . The detailed justification in the analytic audit does not move a contour through a presumed zero. This contradicts the definition of β_* and proves (2.6).

Starting with the source theorem $\beta_* \leq 7/8$, successive applications give $\beta_* \leq f^{\circ n}(7/8)$ for every finite n . The algebraic contraction shows that these numbers decrease to σ_* . Hence $\beta_* \leq \sigma_*$. No common analytic constant across infinitely many iterations is required. Equivalently, if $\beta_* > \sigma_*$, apply the same proof directly with $\sigma = \beta_*$, since $f(\beta_*) < \beta_*$. The source transfer proposition now gives the Dirichlet conclusion in Theorem 1.1. This completes the source-relative proof. $\square$

4 Scope of the refinement

The exact κ -dependent selected-prime capacity, the perturbed low estimate, the extended local Euler region, and the global endpoint inequality are all required for Proposition 2.1. Optimizing the numerical endpoint alone would not yield a zero-free improvement: in particular, the positive-part error for rescaled subsets in the low row estimate must be accounted for before summing the selected-prime tuples.

The exponent-ledger comparison in Appendix B is an optimality statement only within the particular moment bounds and probe family used here. It does not preclude stronger zero-free regions from new arithmetic estimates, a different compensated probe, or a different treatment of the high-frequency terms. None of the deep estimates imported from [1] is independently established in this note.

A Fourth-moment extension and selected-row estimate

The assertions in this section are extensions of specified results of the source manuscript, not new proofs of its finite-correlation identities or its unmarked inverse moment. We use those source results as mathematical inputs. The dependence on the changed parameter is checked below by explicit inequalities; no assertion follows merely by continuity in that parameter.

A.1 Source hypotheses and notation

Retain the arithmetic data, row families, natural zero extensions, fixed finite group Θ , plain polynomials $S_{\psi_k}(n; W)$, and selected prime polynomials of source Lemma `lem:plain`. Write M for its effective row width, z for the total length of the selected prime factors, and $A = n_1 + n_2 + z$. Positive prime-factor coefficients have the fixed finite-ray expansion required in that lemma; no row-dependent coefficient is admitted. Besides the parameter-independent calculations in the proof of `lem:plain`, we use the global logarithmic-derivative estimate of `lem:logarithmic-control`, the strict marked inverse moment `lem:marked`, sixth-power amplification `lem:inverse-amplification`, and the two witnesses in `prop:detector-witness`.

Lemma A.1 (Extended plain fourth moment). *Source Lemma `lem:plain`, with all its coefficient, support, uniformity, and row-family hypotheses, remains valid for*

$$\frac{13}{18} \leq \kappa \leq 1. \quad (\text{A.1})$$

More explicitly, its conclusion is

$$\sum_{k \in \mathcal{R}_z} |S_{\psi_k}(n_1; W_1) S_{\psi_k}(n_2; W_2) Q|^2 \ll Z^{M+\varepsilon}. \quad (\text{A.2})$$

For $z = 0$ its plain lengths are unrestricted in the prescribed bounded ranges. For $z > 0$ assume

$$A + (6\kappa - 1)z = n_1 + n_2 + 6\kappa z \leq M, \quad (\text{A.3})$$

and, when $\kappa < 1$, assume $\beta_* \leq (1 + \kappa)/2$. The slot mesh, exponent losses, and finite semi-norm/height assertions can be chosen uniformly for (A.1).

Proof. There are four substantive uses of κ in the source proof and one uniform Lipschitz bound. We verify each, retaining the source induction and its numerical loss budget.

(i) Put $s_\kappa = (1 + \kappa)/2$. The global assertion of source `lem:logarithmic-control` applies for every fixed $b \geq \beta_*$, not only for $b \geq 7/8$. Thus, on $\Re s = s_\kappa + e$,

$$\frac{L'}{L}(s, \psi) \ll_e \log\{2Q_\psi(3 + |\Im s|)^2\}$$

for each nonprincipal primitive inducing character. The exclusion of the inducing characters in Θ ensures that twisting by each fixed prime-factor coefficient character is still nonprincipal. Mellin inversion and contour shifting give the source prime-factor estimate

$$|Q_\omega|^2 \ll Z^{\kappa z + \varepsilon_1} p_j(\mathbf{W})^b (1 + |\omega|)^h. \quad (\text{A.4})$$

The exponent is $2s_\kappa - 1 = \kappa$. Prime powers contribute only $P^{1/2+\varepsilon_1}$ before central normalization. Contour displacement costs $2ez$, and z stays in a bounded range. On the new compact range, $s_\kappa \geq 31/36 > 1/2$, so the source proof supplies the same uniform choice of e , losses, and fixed height orders. For $\kappa = 1$ use absolute prime counting, exactly as in the source.

(ii) At the width floor, $A \geq z$ and (A.3) imply

$$z \leq \frac{M}{6\kappa} \leq \frac{3M}{13}, \quad \kappa z \leq \frac{M}{6}. \quad (\text{A.5})$$

The source displays $z \leq 2M/9$ at this point, but its terminal exponent uses only the second inequality. Therefore its additional terminal loss remains $\rho/6$ for a width floor $M \leq \rho$.

(iii) In the centered comparison, $A > 5M/6$ and (A.3) imply

$$(6\kappa - 1)z \leq M - A < M/6.$$

Since $6\kappa - 1 \geq 10/3$, this gives the strict bound $z < M/20$. This replaces the source's intermediate inequality $z < M/21 < M/20$ and is exactly the bound used afterwards. Its reflected comparison therefore still satisfies

$$A_{\text{comp}} \leq \frac{3}{2}M - A + 2z + \xi \leq \frac{23}{30}M + \xi, \quad (\text{A.6})$$

$$A_{\text{comp}} + (6\kappa - 1)z \leq \frac{5}{2}M - 2A + 2z + \xi \leq \frac{14}{15}M + \xi. \quad (\text{A.7})$$

Also $A - z > 5M/6 - M/20 > M/2$, as required for the two comparison lengths. Both relevant boundaries retain the source margin $M/15$ before ξ ; its choice $\xi \leq \rho/30$ therefore works unchanged.

(iv) At a smaller-width child define its positive affine defect by

$$F_{\text{act}} = (A_{\text{act}} - M'_{\text{act}} + (6\kappa - 1)z_{\text{act}})_+.$$

Removing a whole prime factor of length d decreases this affine quantity by $6\kappa d$ and, by (A.4), costs κd in the squared exponent. The first prefix reaching $F_{\text{act}}/(6\kappa)$ overshoots by at most the mesh η . If no such prefix exists, all slots are removed and the unrestricted zero-slot child is used. Consequently

$$\kappa d_z \leq F_{\text{act}}/6 + \kappa\eta \leq F_{\text{act}}/6 + \eta. \quad (\text{A.8})$$

This is the source's exact charge, with the same numerical constant $1/6$ and the same single overshoot. No lower bound $\kappa \geq 3/4$ is used here.

(v) The source aggregate boundary-error estimate uses only $0 \leq 6\kappa - 1 \leq 5$, through

$$|\Delta M| + |\Delta A| + 5|\Delta z|.$$

It remains valid. All other transformations in the source proof are independent of κ : fixed-mask removal, reflection of at most two plain factors, finite Gauss identities, the two Poisson steps, the exceptional-character count, the equal-product-scale centered coefficient identity, and the width decrease. In particular the new range changes neither the depth bound nor the source terminal budget $\rho + \delta + \rho/6 + 3\sigma + 5\sigma/3$. The same choices of ξ, η, ε_0 then close its finite induction. This proves (A.2) under the stated source inputs. $\square$

A.2 The selected-row bound

For the rest of this section assume

$$\frac{13}{18} \leq \kappa \leq \frac{3}{4}, \quad \beta_* \leq \frac{1+\kappa}{2}, \quad \alpha = \frac{5}{6}. \quad (\text{A.9})$$

This includes the narrower interval $[7499/10000, 3/4]$ used in the bootstrap. Consider one retained dynamic and amplitude bin of sixth-power-free physical rows $q_u \asymp U$. Require the source's actual witnesses, so $a > 51/100$, and put

$$\delta = 2a - 1 \leq \kappa, \quad 0 \leq q \leq \delta/2, \quad x = q/\delta, \quad c = \frac{1}{3\kappa}.$$

The equality $\kappa = 2\beta_* - 1$ is unnecessary: the displayed upper bound on β_* gives both the positive-slot moment and $\delta \leq \kappa$. Define

$$D_x = 3 - (1 + 2c)x, \quad P_x = 2(1 - cx)(1 - x). \quad (\text{A.10})$$

If the available prime length exceeds $7/37$ by a fixed positive amount, then for every $1 \leq t \leq 3/2$ the source selection mechanism gives

$$\#\mathcal{B} \ll U^{\max\{R_{\text{short}}(t), L(t)\} + \varepsilon} (1 + T_1)^{A_A}, \quad (\text{A.11})$$

where

$$R_{\text{short}}(t) = 1 - \delta + \frac{\delta P_x}{D_x}(3/2 - t), \quad (\text{A.12})$$

$$L(t) = 1 - \delta + (\alpha - \delta)(t - 1). \quad (\text{A.13})$$

There is no additional $\Delta/4$ loss.

Proof of the selected-row bound. The source witness supplies $r + m \geq t - O(\varepsilon)$ and the individual spikes $|M_r|^2 \gg U^{\delta r - \varepsilon}$, $|S_m|^2 \gg U^{\delta m - \varepsilon}$. Slot selection and the moment estimates give respectively

$$U^{1 - \delta r - 2qz + \varepsilon}, \quad U^{1 - 2\delta m - 2qz + \varepsilon}$$

as row-count bounds, with the common height factor understood. The second uses two copies of the plain witness. As in the source, finitely many presentations and dyadic pairs are separated first; rowwise smooth parameters are handled by the common Sobolev argument. These steps and the coefficient-class verification do not depend on κ .

For $r < 1$ and $m < 1/2$, use the exact capacities

$$z_M(r) = (1 - r)/2, \quad z_P(m) = \frac{1 - 2m}{6\kappa} = \frac{c}{2}(1 - 2m).$$

The ideal comparison lines are therefore

$$A_I(r) = 1 - \delta\{x + (1 - x)r\}, \quad (\text{A.14})$$

$$S_t(r) = 1 - \delta\{cx + 2(1 - cx)(t - r)\}. \quad (\text{A.15})$$

The actual plain exponent decreases with m , because its derivative is $-2\delta(1 - cx) < 0$. Thus replacing m by $t - r - O(\varepsilon)$ is legitimate. The two lines cross at

$$r_*(t) = \frac{2(1 - cx)t - (1 - c)x}{D_x}. \quad (\text{A.16})$$

Here $4/9 \leq c \leq 6/13$, and $D_x \geq 53/26 > 0$. At the endpoints,

$$r_*(3/2) = 1, \quad r_*(1) = \frac{2 - (1 + c)x}{3 - (1 + 2c)x}.$$

The derivative of the last quotient with respect to x has numerator $c - 1 < 0$, so

$$r_*(t) \geq r_*(1) \geq \frac{3 - c}{5 - 2c} = \frac{9\kappa - 1}{15\kappa - 2} \geq \frac{23}{37}.$$

For the final inequality, the derivative of $(9\kappa - 1)/(15\kappa - 2)$ is $-3/(15\kappa - 2)^2$, and $\kappa \leq 3/4$. Also

$$t - r_*(t) = \frac{(1 - x)t + (1 - c)x}{D_x}, \quad \frac{1}{3} \leq t - r_*(t) \leq \frac{1}{2}.$$

Indeed this expression increases with t , equals $1/2$ at $t = 3/2$, and its value at $t = 1$ minus $1/3$ is $(1 - c)x/(3D_x) \geq 0$.

Choose the plain count for $r \leq r_*(t)$ and the inverse count for $r \geq r_*(t)$. On the inverse side, requesting $z = z_M(r) - \nu_0$ gives

$$1 - r - 2z = 2\nu_0, \quad 3 - 2r - 8z = 2r - 1 + 8\nu_0 \geq 9/37 + 8\nu_0.$$

These are the two strict marked-moment margins. Its requested capacity is at most $7/37$. On the plain side $m \geq 1/3 - O(\varepsilon)$, so

$$z_P(m) \leq \frac{1}{18\kappa} + O(\varepsilon) \leq \frac{1}{13} + O(\varepsilon) < \frac{7}{37}$$

after reducing preliminary losses. A fixed supply margin therefore suffices uniformly. Decreasing capacities by ν_0 , selecting whole slots from a sufficiently fine mesh, and sorting positive amplitude slots exactly as in the source cost only $O(\varepsilon)$.

The weighted average

$$\frac{2(1 - cx)A_I(r) + (1 - x)S_t(r)}{D_x}$$

is independent of r and equals (A.12). Since A_I decreases and S_t increases, the selected short count is bounded by this common crossing value. No approximation to $1/(6\kappa)$ has been made, so the source comparison loss `eq:capacity-comparison` is absent.

For $r \geq 1$, sixth-power amplification gives $1 - \alpha + (\alpha - \delta)r \leq L(t) + O(\varepsilon)$ since $\delta \leq \kappa < \alpha$ and $r \leq t + O(\varepsilon)$. For $m \geq 1/2$, the unweighted plain moment gives $1 - \delta$. When a capacity is at most ν_0 , the corresponding unweighted moment gives $1 - \delta + O(\nu_0)$: explicitly $2\delta\nu_0$ on the inverse side and at most $6\delta\nu_0$ on the plain side. This covers zero capacities without invoking a marked moment with a shrinking strict margin. All these bounds prove (A.11).

Balancing (A.12) and (A.13) gives

$$t = 1 + \frac{\delta P_x}{2\{(\alpha - \delta)D_x + \delta P_x\}} \in [1, 3/2] \quad (\text{A.17})$$

and therefore the exact envelope

$$R_\kappa(\delta, x) = 1 - \delta + \frac{(\alpha - \delta)\delta P_x}{2\{(\alpha - \delta)D_x + \delta P_x\}}. \quad (\text{A.18})$$

With no selected slots, setting $t = 1$ still gives the source exponent $1 - 2\delta/3$, without any supply assumption.

Scope of the extension. The argument proves the changed moment and selected-row interfaces from the named source inputs. The source witness and inverse-amplification statements contain no $7/8$ lower threshold; their relevant restrictions are respectively $a > 51/100$ and the sixth-power-free row condition. The only zero-free assertion used in the changed fourth moment is $\beta_* \leq (1 + \kappa)/2$, explicitly stated above. The analytic continuation regions for the physical compensated probe, the changed geometry, and the global endpoint inequality are separate interfaces and must be verified elsewhere before a new half-plane theorem is concluded. This section makes no unsupported extension of those interfaces.

B Endpoint optimization and the bootstrap

This section is an algebraic implication from the row-count and exponent interfaces stated in the main text. In particular, its use for zero-free regions requires those interfaces at every input parameter used below. Put

$$\alpha = \frac{5}{6}, \quad I_\kappa = \left[\frac{7499}{10000}, \frac{3}{4} \right].$$

Every terminating decimal used below is an exact rational number.

B.1 The generalized row envelope

For $\kappa \in I_\kappa$, $0 \leq x \leq 1/2$, define

$$D_\kappa(x) = 3 - \left(1 + \frac{2}{3\kappa}\right)x, \quad P_\kappa(x) = \left(2 - \frac{2x}{3\kappa}\right)(1 - x).$$

The selected inverse and plain count exponents are, respectively,

$$A_I(r) = 1 - \delta\{x + (1 - x)r\}, \quad S_t(r) = 1 - \delta\left\{\frac{x}{3\kappa} + \left(2 - \frac{2x}{3\kappa}\right)(t - r)\right\}.$$

Their intersection gives

$$R_{\text{short}}(t) = 1 - \delta + \frac{\delta P_\kappa(x)}{D_\kappa(x)}(3/2 - t), \quad L(t) = 1 - \delta + (\alpha - \delta)(t - 1).$$

For $0 < \delta \leq \kappa$, let

$$\mathcal{J} = (\alpha - \delta)D_\kappa(x) + \delta P_\kappa(x), \quad t_* = 1 + \frac{\delta P_\kappa(x)}{2\mathcal{J}}, \quad v = \frac{(\alpha - \delta)\delta P_\kappa(x)}{2\mathcal{J}}.$$

Then $1 < t_* < 3/2$, and exact substitution gives

$$R_{\text{short}}(t_*) = L(t_*) = 1 - \delta + v.$$

This is the minimum of the maximum of these two affine count bounds.

The supply margins also persist on I_κ . Writing $c = 1/(3\kappa)$, the crossing of A_I and S_t occurs at

$$r_*(t) = \frac{(2 - 2cx)t - (1 - c)x}{3 - (1 + 2c)x}, \quad t - r_*(t) = \frac{(1 - x)t + (1 - c)x}{3 - (1 + 2c)x}.$$

For $1 \leq t \leq 3/2$, one has

$$r_*(t) \geq r_*(1) \geq \frac{3 - c}{5 - 2c} \geq \frac{23}{37}, \quad \frac{1}{3} \leq t - r_*(t) \leq \frac{1}{2}.$$

Here $r_*(1)$ decreases with x , whereas $(3 - c)/(5 - 2c)$ increases with $c \geq 4/9$. The analogous assertion for $t - r_*(t)$ follows by differentiating $(1 - cx)/(3 - (1 + 2c)x)$ with respect to x . Consequently the inverse capacity is at most $7/37$, its second strict width margin is at least $9/37$, and the plain capacity is at most $1/(18\kappa) < 7/37$.

B.2 The optimized geometry

For each $\kappa \in I_\kappa$, let d_κ be the smaller root of

$$p_\kappa(d) := (216\kappa - 18)d^2 - (468\kappa - 57)d + 150\kappa - 20 = 0. \quad (\text{B.1})$$

Define

$$v_\kappa(d) = \frac{(6\kappa - 1)d(5/6 - d)}{2\{(5/6)(15\kappa - 2) + (1 - 9\kappa)d\}},$$

and set

$$\ell_\kappa = \frac{5 - 6d_\kappa}{9 + 18d_\kappa}, \quad b_\kappa = \frac{(1 + 3\ell_\kappa)v'_\kappa(d_\kappa)}{1 - v'_\kappa(d_\kappa)}. \quad (\text{B.2})$$

The remaining geometry is

$$l_x = \frac{1 - \ell - b}{2}, \quad l_y = \frac{1 - \ell + b}{2}, \quad h = \frac{1 + 3\ell + b}{2}, \quad C = -\frac{1}{4} + \frac{5}{4}\ell + \frac{b}{6},$$

where initially $(\ell, b) = (\ell_\kappa, b_\kappa)$. With $y = 1/2 - x$, the central endpoint exponent is

$$E(\delta, y) = hv + C - \delta(\ell y + b/2). \quad (\text{B.3})$$

At $y = 0$, the quantity v equals $v_\kappa(\delta)$.

Here are exact rational enclosures used in the proof. Put $d_- = 19429/50000$ and $d_+ = 38859/100000$. Since p_κ is affine in κ , the four signs

	$\kappa = 7499/10000$	$\kappa = 3/4$
$p_\kappa(d_-)$	1875081893/3125000000000	105619/156250000
$p_\kappa(d_+)$	-15256860787/12500000000000	-715571/625000000

give a root in (d_-, d_+) for every $\kappa \in I_\kappa$. The derivative in this rectangle is negative, between -183 and -182 , so this is the smaller root and it is unique there. In particular,

$$0.38858 < d_\kappa < 0.38859, \quad 0.1668 < \ell_\kappa < 0.1669, \quad 0.1233 < b_\kappa < 0.1236. \quad (\text{B.4})$$

For completeness, the intermediate rational interval bounds are

$$0.1668348 < \ell_\kappa < 0.1668405, \quad 0.075935 < v'_\kappa(d_\kappa) < 0.076048.$$

They follow by substituting $d_- \leq d \leq d_+$ and $\kappa \in I_\kappa$ into

$$v'_\kappa(d) = \frac{A\{(5/6)B(5/6 - 2d) + (B - A)d^2\}}{2\{(5/6)B + (A - B)d\}^2}, \quad A = 6\kappa - 1, \quad B = 15\kappa - 2.$$

The last displayed bounds, substituted in (B.2), give the asserted interval for b . All denominators in these substitutions are positive.

The optimizing identities are exact. If $W = (5/6)(15\kappa - 2) + (1 - 9\kappa)d$, then

$$36W\{v_\kappa(d) - d + 1/3\} = p_\kappa(d).$$

Thus $v_\kappa(d_\kappa) = d_\kappa - 1/3$. The definitions of ℓ_κ and b_κ now imply

$$E(d_\kappa, 0) = 0, \quad \partial_\delta E(d_\kappa, 0) = 0. \quad (\text{B.5})$$

B.3 The global endpoint inequality

We prove

$$E(\delta, y) \leq 0 \quad (0 \leq \delta \leq \kappa, \quad 0 \leq y \leq 1/2), \quad (\text{B.6})$$

with equality only at $(\delta, y) = (d_\kappa, 0)$. In fact the numerator argument below works for every real δ .

Put $c = 1/(3\kappa)$ and write

$$\begin{aligned} j_0(y) &= (225 - 90c) + (90 + 180c)y, \\ j_1(y) &= (-162 + 54c) + (108 - 216c)y + 216cy^2, \\ P(y) &= 1 - c/2 + 2y + 2cy^2, \quad k(y) = b/2 + \ell y. \end{aligned}$$

Then $j = j_0 + \delta j_1 = 108\mathcal{J}$, and multiplication of (B.3) gives the exact identity

$$-jE = a(y)\delta^2 + B(y)\delta + C_0(y), \quad (\text{B.7})$$

where

$$a = kj_1 + 54hP, \quad B = kj_0 - Cj_1 - 45hP, \quad C_0 = -Cj_0.$$

Write $a = \sum_{i=0}^3 a_i y^i$, $B = \sum_{i=0}^2 B_i y^i$, and $C_0 = C_0^{(0)} + C_0^{(1)} y$. The following table is a rational interval certificate over the entire box

$$\ell \in [0.1668, 0.1669], \quad b \in [0.1233, 0.1236], \quad c \in [4/9, 10000/22497].$$

It is obtained directly from the preceding polynomial identities by interval addition and multiplication, not by sampling parameter values.

	0	1	2	3
a_i	[25.56, 25.61]	[65.38, 65.44]	[46.88, 46.93]	[16.01, 16.03]
B_i	[-19.92, -19.84]	[-31.51, -31.43]	[-2.14, -2.08]	
$C_0^{(i)}$	[3.84, 3.88]	[3.53, 3.57]		

To make the certificate reproducible without elimination, set

$$u_0 = 225 - 90c, \quad u_1 = 90 + 180c, \quad w_0 = -162 + 54c, \quad w_1 = 108 - 216c, \quad w_2 = 216c.$$

The individual coefficients to which the interval operations apply are

$$\begin{aligned} a_0 &= (b/2)w_0 + 54h(1 - c/2), & a_1 &= (b/2)w_1 + \ell w_0 + 108h, \\ a_2 &= (b/2)w_2 + \ell w_1 + 108hc, & a_3 &= \ell w_2, \\ B_0 &= (b/2)u_0 - Cw_0 - 45h(1 - c/2), & B_1 &= (b/2)u_1 + \ell u_0 - Cw_1 - 90h, \\ B_2 &= \ell u_1 - Cw_2 - 90hc, & C_0^{(0)} &= -Cu_0, \quad C_0^{(1)} = -Cu_1. \end{aligned}$$

The constant coefficient of $4aC_0 - B^2$ is zero by (B.5): the quadratic at $y = 0$ has the double root d_κ . Its coefficients of y, y^2, y^3, y^4 are bounded below by, respectively,

$$109.7856, \quad 565.1047, \quad 772.9964, \quad 221.4816.$$

For example, the first lower bound is exactly

$$4(25.56 \cdot 3.53 + 65.38 \cdot 3.84) - 2 \cdot 19.92 \cdot 31.51 = 109.7856.$$

The remaining three follow from

$$\begin{aligned} [y^2](4aC_0 - B^2) &= 4(a_1C_0^{(1)} + a_2C_0^{(0)}) - B_1^2 - 2B_0B_2, \\ [y^3](4aC_0 - B^2) &= 4(a_2C_0^{(1)} + a_3C_0^{(0)}) - 2B_1B_2, \\ [y^4](4aC_0 - B^2) &= 4a_3C_0^{(1)} - B_2^2. \end{aligned}$$

Thus for $y > 0$, $a(y) > 0$ and $4a(y)C_0(y) - B(y)^2 > 0$. Completion of the square proves that the right side of (B.7) is strictly positive. At $y = 0$ it equals $a_0(\delta - d_\kappa)^2$. Finally $j > 0$ on the relevant domain: in the y -coordinates, $D = (5/2 - c) + (1 + 2c)y > 2$, $P > 3/4$, $D - P \geq 1$, and

$$\frac{35}{48} < \mathcal{J} \leq \frac{5}{2} \quad (0 \leq \delta \leq \kappa \leq 3/4).$$

This proves (B.6) and its equality statement.

B.4 Strict approximations and the remaining algebraic margins

Hold $\kappa, b = b_\kappa$ fixed and replace ℓ_κ by $\ell_\kappa - \varepsilon$, with $0 < \varepsilon \leq 10^{-6}$. Since $v \geq 0$,

$$\frac{\partial E}{\partial \ell} = \frac{3}{2}v + \frac{5}{4} - \delta y \geq \frac{5}{6} \quad (0 \leq \delta \leq 5/6, \quad 0 \leq y \leq 1/2).$$

Consequently the perturbed geometry has the uniform strict margin

$$E \leq -\frac{5}{6}\varepsilon. \tag{B.8}$$

Its output exponent is

$$\sigma_0 = \frac{11}{12} - \frac{\ell}{4} = 1 - \frac{2}{9 + 18d_\kappa} + \frac{\varepsilon}{4}.$$

The entire perturbed family remains in the parameter box above. That box also certifies

$$\begin{aligned} 0.35475 \leq l_x \leq 0.35495, \quad 0.4782 \leq l_y \leq 0.4784, \quad 0.81185 \leq h \leq 0.81215, \\ \frac{\ell}{h} - \frac{7}{37} \geq \frac{263}{16243} > 0, \quad l_y - \ell - \frac{11b}{6} \geq \frac{1697}{20000} > 0. \end{aligned}$$

For the floor bin $\delta_0 = 1/50$, the ideal bound $R = 1$ and $q \leq \delta_0/2$ give

$$E_{\text{floor}} \leq C + \frac{\delta_0}{2}(1 + 3\ell) \leq -\frac{721}{125000} < 0.$$

The small-row inequality specified in the exponent interface obeys

$$h\left(\frac{17}{50} - \frac{1}{6}\right) - \frac{l_y}{2} + \frac{1}{50} \leq -\frac{117491}{1500000} < 0.$$

These are uniform strict algebraic margins. They do not replace the analytic hypotheses behind the corresponding source estimates.

B.5 The bootstrap map and its fixed point

For an input boundary $\sigma \in I_\sigma = [17499/20000, 7/8]$, put $\kappa = 2\sigma - 1$ and define

$$f(\sigma) = 1 - \frac{2}{9 + 18d_{2\sigma-1}}. \quad (\text{B.9})$$

Assuming the analytic interfaces, the strict geometries above prove every output boundary $f(\sigma) + \varepsilon/4$, and hence, by taking their intersection, the boundary $f(\sigma)$ itself. This limiting step requires no estimate uniform as $\varepsilon \rightarrow 0$: any fixed putative zero to the right is excluded by one sufficiently small positive ε .

The root enclosures give

$$\frac{349861}{399861} < f(\sigma) < \frac{699731}{799731},$$

so f maps I_σ strictly into itself. Implicit differentiation of (B.1) gives

$$d'_\kappa = -\frac{216d_\kappa^2 - 468d_\kappa + 150}{2(216\kappa - 18)d_\kappa - (468\kappa - 57)}.$$

On the root rectangle the numerator is between $3/4$ and $4/5$, and the denominator is between -183 and -182 . Thus $0 < d'_\kappa < 1/200$, and

$$0 < f'(\sigma) = \frac{72d'_{2\sigma-1}}{(9 + 18d_{2\sigma-1})^2} < \frac{1}{500}.$$

It follows that f has a unique fixed point σ_* in I_σ . The iterates beginning at $\sigma_0 = 7/8$ decrease strictly to σ_* . Indeed $f(\sigma) - \sigma$ is strictly decreasing and changes sign in the interval, and monotonicity of f keeps every iterate above its fixed point. In particular,

$$f(\sigma) < \sigma \quad \text{if and only if} \quad \sigma \in (\sigma_*, 7/8].$$

Thus, if the analytic argument is formulated using a putative boundary $\beta_* \in (\sigma_*, 7/8]$, one may apply the geometry directly at $\kappa = 2\beta_* - 1$ and choose $0 < \varepsilon < 4\{\beta_* - f(\beta_*)\}$. This gives a strictly smaller output boundary and a strictly negative endpoint margin in one step; an infinite analytic iteration is not needed for that formulation.

At a fixed point,

$$\kappa = 2\sigma_* - 1 = \frac{18d + 5}{18d + 9}.$$

Substitution into (B.1) yields

$$p_{(18d+5)/(18d+9)}(d) = \frac{3(1188d^3 - 2160d^2 + 171d + 190)}{18d + 9}.$$

Consequently the exact limiting boundary is

$$\boxed{\sigma_* = 1 - \frac{2}{9 + 18d_*}, \quad 1188d_*^3 - 2160d_*^2 + 171d_* + 190 = 0, \quad 0.38858 < d_* < 0.38859.} \quad (\text{B.10})$$

The cubic is strictly decreasing on that interval. Exact substitution at the rational endpoints 0.38858335426603 and 0.38858335426604 gives opposite signs, and hence

$$\sigma_* \approx 0.87495701942009895.$$

Since $d \mapsto 1 - 2/(9 + 18d)$ is increasing and

$$\frac{162861}{186136} = 1 - \frac{2}{9 + 18 \cdot \frac{162797}{418950}}, \quad \frac{162797}{418950} > 0.38858335426604 > d_*,$$

the rational bound (1.2) follows; its margin is about $1.25 \cdot 10^{-9}$. If each analytic bootstrap implication holds, every zero lies to the left of every iterate and therefore to the left of σ_* .

B.6 Optimality within the given exponent bounds

The following final observation is conditional on the broader low-frequency penalty formula in the main text. It is an optimality statement about the existing bounds, not an obstruction to improved arithmetic estimates.

Allow a nonnegative geometry deficit $\omega = 1 - M - \ell$, and put

$$p = \frac{(5\ell + 2\omega - 1)_+}{8}, \quad \sigma_0 = \frac{11}{12} - \frac{\ell}{4} - \frac{\omega}{12} + p, \quad h = \frac{1 + 3\ell + b + \omega}{2}.$$

Suppose the high-frequency exponent is

$$E = -\frac{1}{4} + \frac{5}{4}\ell + \frac{3}{4}\omega + \frac{b}{6} - p - \delta(\ell y + b/2) + hv.$$

At the same critical point $(\delta, y) = (d_\kappa, 0)$, the identity $v = d_\kappa - 1/3$ makes the coefficient of b vanish exactly. The exponent there is

$$E = \left(\frac{3}{4} + \frac{3}{2}d_\kappa\right)(\ell - \ell_\kappa) + \left(\frac{7}{12} + \frac{1}{2}d_\kappa\right)\omega - p.$$

Therefore the necessary condition $E \leq 0$ implies

$$\boxed{\sigma_0 - f(\sigma) \geq \frac{\omega}{9 + 18d_\kappa} + p \left(1 - \frac{1}{3 + 6d_\kappa}\right) \geq 0.}$$

Thus allowing $M + \ell < 1$, changing b , or activating this low-frequency penalty cannot beat the optimized map while these exponent bounds remain unchanged. A better boundary requires a new estimate or an observable with a different exponent ledger.

C Analytic estimates for the perturbed probe

Throughout this section, $F = \mathbb{Q}(\sqrt{-3})$, and β_* denotes the supremum in `old-eq:1.1b` of the source manuscript: the supremum of the real parts of nontrivial zeros of *all primitive finite-order Hecke L-functions over F*, with $1/2$ adjoined. Poles are excluded. The supremum must not be replaced by the supremum for ζ alone: the reflected numerator and denominator characters belong to the larger Hecke family.

We use the exact finite probe of `eq:compensated-probe-definition`, with the same slot operation, zero extensions, excluded primes, and coefficient conventions, but with variable real geometry

$$l_y = l_x + b, \quad M = 2l_x + b, \quad h = 1 - l_x + \ell, \quad C(s) = s + \frac{l_x}{2} - 1 + \frac{h}{6}.$$

The slot lengths are positive and sum to ℓ . Define

$$\sigma_0 = 1 - \frac{h}{6} + \frac{b}{12}, \quad C(\sigma_0) = \frac{l_x}{2} + \frac{b}{12}.$$

Here M is a real log-length, not an inverse polynomial.

C.1 Explicit sufficient geometric conditions

The analytic arguments below require

$$\begin{aligned} b > 0, \quad l_x > \ell > 0, \quad h > \frac{1}{2}, \quad M + \ell \leq 1, \quad 1 + 3\ell - 2M \leq 0, \\ l_y - \ell - \frac{11b}{6} > 0, \quad \sigma_0 > \frac{87}{100}. \end{aligned} \tag{C.1}$$

In addition, the central arithmetic estimate needs a fixed prime-supply margin

$$\frac{\ell}{h + \zeta} > \frac{7}{37} \tag{C.2}$$

for a sufficiently small positive ζ . This condition is used only for $1/2 \leq d \leq h + \zeta$. Shorter row ranges use the unselected witness estimate.

For clarity, the arithmetic inputs to this section are separate from these geometric conditions. They are the generalized fourth-moment lemma with its stated zero-free input, the corresponding selected-witness row count, and the endpoint inequality described below. The source states its fourth-moment lemma only for $\kappa \geq 3/4$; using $\kappa < 3/4$ requires the extension proved in Section A. The present section does not assume that extension merely by changing a parameter in the source statement.

C.2 The low estimate with variable geometry

Fix a rescaled subset of slots of total length d , and put

$$0 \leq d \leq \ell, \quad M' = M - 2d, \quad \ell' = \ell - d.$$

The proof of `lem:probe-row-norm`, before its last substitution of the numerical geometry, gives

$$\sum_{0 < q_m \ll Z^{M'}} |B_{m,\sigma}^J(Z)|^2 \ll Z^{M' + (1 + 3\ell - 2M + d)_+ / 4 + \epsilon}. \tag{C.3}$$

Here is why its earlier use of the geometry remains valid. In the notation of that proof, $H \leq M' - O + \epsilon_Z$ and $2A_0 \leq O + \epsilon_Z$. The dual-length calculation becomes

$$T_d - H \leq M' + \ell' - 1 + O(\epsilon_Z + |\theta_N|) = M + \ell - 1 - 3d + O(\epsilon_Z + |\theta_N|) \leq -3d + O(\epsilon_Z + |\theta_N|).$$

It follows, exactly as in the source, that the maximum involving the residual row length is $H + O(\epsilon)$. The two subsequent branches of `old-eq:5.3` bound the energy by

$$M' - \Delta_H + O(\epsilon), \quad M' + \frac{1 + 3\ell' - 2M' - 2\Delta_H}{4} + O(\epsilon).$$

Since $\Delta_H \geq -O(\epsilon)$, these yield (C.3). No assumption $M' \geq 1/2$ enters this calculation. Positivity of M' , which is all the later scale argument needs, follows from $l_x > \ell$ and $b > 0$.

For this subset, the source's additive Gram estimate `prop:probe-gram` has

$$P_a = \frac{Y'^2}{Q} = q_{b_*}^{-1} Z^b.$$

Thus $P_a \geq 1$ eventually. The strict inequality in (C.1) involving l_y gives $Y'/P_a^{11/6} \gg Z^{l_y - \ell - 11b/6}$, uniformly in the fixed rescaled tuple. Consequently the nonexceptional Gram term P_a^2/Y' is bounded by a constant times $P_a^{1/6}$. Combining the Gram bound with (C.3), then summing the rescaled prime tuples as in `old-eq:5.5`, gives the additional exponent

$$-d + \frac{(1 + 3\ell - 2M + d)_+}{8} \leq 0.$$

Indeed, if the positive part is nonzero this equals $(1 + 3\ell - 2M - 7d)/8 \leq 0$; otherwise it is $-d$. It follows that the same independently defined physical probe satisfies

$$|I_{\eta, \text{modified}}(Z)| \ll_{\eta, \epsilon} Z^{l_x/2 + b/12 + \epsilon} = Z^{C(\sigma_0) + \epsilon}. \quad (\text{C.4})$$

More generally, without the condition $1 + 3\ell - 2M \leq 0$, but keeping $M + \ell \leq 1$ and the other scale and Gram conditions, the same proof gives the additional low exponent

$$p = \frac{(1 + 3\ell - 2M)_+}{8}.$$

Indeed $-d + (1 + 3\ell - 2M + d)_+/8$ is nonincreasing in d : its two possible slopes are -1 and $-7/8$. Its maximum at $d = 0$ is exactly p . If $\mathfrak{d} = 1 - M - \ell \geq 0$, the resulting low threshold and central endpoint are consequently

$$\sigma_0 = \frac{11}{12} - \frac{\ell}{4} - \frac{\mathfrak{d}}{12} + p, \quad p = \frac{(5\ell + 2\mathfrak{d} - 1)_+}{8},$$

$$E(h) = -\frac{1}{4} + \frac{5\ell}{4} + \frac{3\mathfrak{d}}{4} + \frac{b}{6} - p - \delta(\ell y + b/2) + hv, \quad R = 1 - \delta + v.$$

These formulas justify the general low-penalty ledger used in the geometry-optimality argument; they do not assert optimality among different arithmetic moment estimates.

C.3 Extension of the principal Euler region

The displayed local formulas in `lem:local-euler` extend its second region to

$$\Re s \geq \frac{87}{100}, \quad \Re w \geq \frac{19}{20}, \quad \Re z \geq \frac{33}{200}. \quad (\text{C.5})$$

This is a direct estimate, not a continuity assertion. The source's local quantities satisfy

$$|R| \leq Q^{-221/100}, \quad |V| \leq Q^{-99/100}, \quad |D| \leq Q^{-87/100}.$$

Using `eq:local-H-defect` and the displayed table of J_j , the nearest unramified defect is $Q^{1 - \Re s - \Re w - 6\Re z} \leq Q^{-181/100}$, and the nearest ramified defect is $Q^{1 - \Re s - \Re w} \leq Q^{-41/50}$. The other unramified terms are bounded, for example, by $Q^{-182/100}$, $Q^{-186/100}$, or $Q^{-221/100}$. The other ramified boundary terms have exponents at most

$$-111/100, \quad -148/100, \quad -156/100, \quad -193/100, \quad -222/100.$$

Therefore, uniformly in all heights and unit phases,

$$H_p - 1 = O(Q^{-181/100}) \ (p \nmid u), \quad H_p - 1 = O(Q^{-41/50}) \ (p \mid u). \quad (\text{C.6})$$

Normal convergence and the divisor-product argument of the source give $\mathcal{H}_{\eta,u} \ll_{\epsilon} q_u^{\epsilon}$ in (C.5). For $u = 1$, deleting primes up to P_0 gives $\mathcal{H}_{\eta,1} = 1 + O(P_0^{-4/5})$: the stronger tail exponent supplied by (C.6) is $81/100$.

The source's quotient-free formula `old-eq:5.13c` for G_p continues the full selected-tuple correction into this region. There is no division by a possibly vanishing H_p or P_p in that formula. For the principal row, after taking P_0 large enough that $|H_p - 1| < 1/2$, `old-eq:5.13b` gives

$$G_p/H_p = -1 + O(Q^{-87/100}).$$

The contributing error powers are bounded by $Q^{-87/100}, Q^{-99/100}, Q^{-134/100}, Q^{-94/100}$. The fixed-ray prime normalizer `lem:ray-prime-normalizer` therefore proves, for the source's $A_T(Z)$,

$$|A_T(Z)| \asymp_{\eta} (\log Z)^{-K}, \quad \mathfrak{H}_{\eta,1,Z}(s, 1, 1/6) = H_{\eta}(s) Z^{\ell/6} A_T(Z) (1 + O(Z^{-\mu})), \quad (\text{C.7})$$

uniformly on $\Re s = \beta_* + e$, for any fixed $0 < \mu < (87/100) \min_i \ell_i$. Here $H_{\eta}(s) = \mathcal{H}_{\eta,1}(s, 1, 1/6)$. Increasing the same fixed P_0 ensures

$$\sup_{\Re s > \sigma_0} |H_{\eta}(s) - 1| \leq \frac{1}{2}.$$

All excluded sets and lower thresholds may depend on the fixed target; the exponent μ is chosen before it. Only a fixed-ray prime asymptotic is used, not a uniform moving-conductor prime theorem.

C.4 The other contours and the exact central exponent

The exact Poisson identity `eq:compensated-poisson-identity` still holds for this physical probe, because at each fixed Z the modification is a finite linear combination of the same absolutely convergent rescaled base identities. Its weight has the general form in `eq:stage-mellin-exponent`. In particular it requires $h = 1 - l_x + \ell$, not the original numerical values of the geometry.

The first Euler region `eq:local-region-one` and the dynamic local-error estimate `prop:probe-errors` are unchanged. The proof of `lem:fixed-bin-contour` uses only this first region, the buffered zero rectangles, and the global line to the right of β_* . Its stated assumption $\sigma_0 \geq 7/8$ is not used elsewhere in that proof. The same is true of the algebraic accounting in `lem:bin-contour-accounting`.

For a bin $a = (1 + \delta)/2$, prime-amplitude mean q , and row-count exponent R , that accounting gives exactly

$$E(d) = a - \sigma_0 + h(17/50 - 1/6) - al_y - \ell/2 + q\ell + d(R + \delta/2 - 17/50). \quad (\text{C.8})$$

At $d = h$, writing $y = 1/2 - q/\delta$, it becomes

$$E(h) = h(R + \delta + 1/2) - 1 - \ell(1 + \delta y) - b(7/12 + \delta/2). \quad (\text{C.9})$$

The generalized arithmetic endpoint lemma is required to prove $E(h) \leq 0$, uniformly for the relevant δ, y . It is sufficient that this inequality be non-strict. Under the contradiction $\Delta = \beta_* - \sigma_0 > 0$, it gives a strict saving Δ relative to $C(\beta_*)$. The extension to $d \leq h + \zeta$, the

slot rounding, the detector padding, and all amplitude losses can be chosen to consume less than $\Delta/2$. For $d \leq h$, the frequency slope is positive whenever $R \geq 1 - \delta$ and $\delta \leq 3/4$, since

$$R + \delta/2 - 17/50 \geq 33/50 - \delta/2 \geq 57/200.$$

The floor $\delta_0 = 1/50$ uses $R = 1$, without a witness. Its endpoint is bounded above by

$$E_{\text{floor}}(h) = h(3/2 + 1/50) - 1 - \ell - b(7/12 + 1/100). \quad (\text{C.10})$$

For $d \leq 1/2$, the unselected estimate $R = 76/75 - 2\delta/3$ is available, including the floor. Using $q \leq \delta/2$, its worst endpoint is the larger value, for $\delta \in [1/50, 3/4]$, of the affine function

$$\frac{1}{2} - \sigma_0 + h(17/50 - 1/6) - \frac{l_y + \ell}{2} + \frac{1}{2}(76/75 - 17/50) + \delta \left(\frac{5}{12} + \frac{\ell - l_y}{2} \right). \quad (\text{C.11})$$

Thus checking its two endpoints is sufficient for the whole short range.

The principal-residue proof `lem:principal-residue-interface` now takes place in (C.5); its two scalar poles and their residues are unchanged. By (C.7), the three resulting errors, after normalization, have exponents

$$C(\beta_*) + (1 + h)e - l_y/20 + \epsilon, \quad C(\beta_*) + e - h/600 + \epsilon, \quad C(\beta_*) + e - \mu + \epsilon.$$

These are strict savings once the positive padding is sufficiently small.

For small nonprincipal rows the source's `lem:compensated-absolute-local-bounds` remains valid with $\Re s \geq 87/100$. Off u , its four error exponents are at most

$$-87/100, \quad -51/25, \quad -239/100, \quad -77/50;$$

on u , the largest term is still $Q^{1/2}$. Hence the positive tuple sum is $O_\epsilon(U^\epsilon Z^{17\ell/50})$. In the source proof of `lem:outer-row-tails`, replace the claimed path in $\mathcal{D}_1(3/8)$ by one in $\mathcal{D}_1(1/3)$. This is legitimate because $\beta_* > \sigma_0 > 87/100 > 5/6$. For $d_{\min} = 1/100$, the resulting small-row exponent relative to $C(\beta_*)$ is

$$h(17/50 - 1/6) - l_y/2 + \frac{63}{5000} + e + \epsilon. \quad (\text{C.12})$$

For large rows, the source's absolute lines $(s, w, z) = (2, 2, v)$, $v > 2$, are unchanged. With $B_0 = l_x/2 + 1 + l_y$, their sum for $U > Z^{h+\zeta}$ is bounded by

$$Z^{B_0 + (h+\zeta)(1+\epsilon) - \zeta v + \epsilon}.$$

A fixed sufficiently large v makes it smaller than any required power. The order is: choose ζ , then v , both before the target and the eventual scale.

C.5 Verification on the proposed geometry interval

The optimized family in the main proof obeys the rational enclosures

$$\frac{16683}{100000} < \ell < \frac{16685}{100000}, \quad \frac{123}{1000} < b < \frac{124}{1000}, \quad M = 1 - \ell, \quad h = \frac{1 + 3\ell + b}{2}.$$

They imply

$$0.811745 < h < 0.812275, \quad 0.354575 < l_x < 0.355085, \quad 0.478075 < l_y < 0.478585,$$

and $\sigma_0 = 11/12 - \ell/4 > 0.8749 > 87/100$. In particular $1 + 3\ell - 2M = 5\ell - 1 < -0.1657$, $l_x - \ell > 0.1877$, $l_y - \ell - 11b/6 > 0.0843$, and $\ell/h > 0.205 > 7/37$. Thus (C.1) and (C.2) hold with uniform positive margins.

For this family (C.10) simplifies to

$$-\frac{6}{25} + \frac{32\ell}{25} + \frac{b}{6} < -0.0057.$$

The coefficient of δ in (C.11) is positive. At $\delta = 3/4$, that expression simplifies to

$$-\frac{871}{1200} + \frac{243h}{200} - \ell - \frac{23b}{24} = -\frac{71}{600} + \frac{329\ell}{400} - \frac{421b}{1200} < -0.0242.$$

Finally (C.12), before its adjustable losses, is less than -0.0856 . The principal geometric margins satisfy $l_y/20 > 0.0239$ and $h/600 > 0.00135$. These inequalities verify every outer and floor interface without an appeal to qualitative robustness. The nonfloor endpoint remains the explicit arithmetic polynomial inequality proved separately.

C.6 Mellin continuation and order of choices

Set $J_\eta = I_{\eta, \text{modified}}/(c_S A_T)$, with the same normalizer in the low and high estimates. The positive scalar c_S is the product of the two principal residues from the source. Equations (C.4) and (C.7) give $J_\eta \ll Z^{C(\sigma_0)+\omega}$ for any fixed $0 < \omega < \Delta$. The central endpoint estimate and the explicit margins above give a target-independent $m > 0$ and

$$|J_\eta - f_\eta| \ll_{\eta, N} Z^{C(\beta_*)-m}(1 + T_1)^{A_\eta} + Z^{B_\eta} T_1^{-N},$$

where A_η, B_η are finite and fixed before the external tail order N . This is exactly the contract in `lem:late-height-closure`. Its proof chooses $T_1 = Z^{\tau_\eta}$ after η , with $\tau_\eta \leq m/[4(A_\eta + 1)]$ and the detector's additional ceiling, and then chooses N . It supplies the common saving $m/2$. The physical probe and its principal signal do not depend on T_1 .

For completeness, the signal is

$$f_\eta(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{C(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s, \eta)} ds.$$

The common low and high savings imply $f_\eta(Z) \ll Z^{C(\beta_*)-\varepsilon_*}$, with $0 < \varepsilon_* < \Delta$ independent of η , after shrinking the saving if necessary. Thus the half-plane to be continued remains inside $\Re s > \sigma_0$, where the multiplier is nonvanishing. Moving only the absolutely convergent line $\Re s = 2$ to arbitrary larger real parts gives arbitrarily rapid power decay as $Z \downarrow 0$. Consequently

$$\int_0^\infty f_\eta(Z) Z^{-C(s)} \frac{dZ}{Z}$$

is holomorphic for $\Re s > \beta_* - \varepsilon_*$. Fourier inversion on $\Re s = 2$, followed by the identity theorem, identifies it with $e^{(s-5/6)^2} H_\eta(s)/L_F^S(s, \eta)$ where $\Re s > 1$. Since $|H_\eta| \geq 1/2$, it continues $1/L_F^S$ holomorphically to the larger half-plane. This contradicts the definition of β_* , exactly as in `lem:continuation-criterion`. The deleted factors have no zeros for $\Re s > 0$.

Thus the analytic interfaces prove the claimed new half-plane once the generalized arithmetic moment, row-count, and nonfloor endpoint lemmas have been established. The transfer to all finite-order Hecke characters and to Dirichlet L -functions is the source's `prop:hecke-dirichlet-transfer`; principal poles at one are allowed. For a limiting bootstrap value, apply this implication at each finite stage and take the infimum of the resulting bounds on the same β_* . One must not claim a fresh strict endpoint saving at the limiting tangency without the contradiction parameter $\Delta > 0$.

AI assistance

The author found this somewhat accidentally. He asked GPT-6 Astra to try to apply the results of OpenAI's $7/8$ proof to some independent investigations of the Riemann hypothesis that he was pursuing. It did the opposite, and applied some optimization ideas he had tried to the OpenAI proof. Somehow, this worked and produced a very slight improvement.

References

- [1] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane $\operatorname{Re} s > 7/8$* , 30 September 2026. Version of 6 October 2026, [source TeX at GitHub \(commit `adc7f124`\)](#).